

RISHABH AGRAHARI

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EDUCATION

Indian Institute of Information Technology, Gwalior

Aug 2023 – Apr 2028

B.Tech + M.Tech in Information Technology

CGPA: 8.26 / 10

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming, Machine Learning

Dr Rizvi Learners Academy, Jaunpur

Apr 2021 – Mar 2022

CBSE Class XII Boards

93.2%

PROJECTS

TrafficAI — Automated Traffic Violation Detection System

🐙 GitHub | June 2026

Python, YOLOv8, OpenCV, EasyOCR, FastAPI, Streamlit, SQLite, Docker

- Built an end-to-end computer vision system detecting, classifying, and documenting **7 types of traffic violations** (helmet, seatbelt, triple riding, wrong-side, stop-line, red-light, illegal parking) with license-plate OCR, processing each image in **~3 s on CPU**.
- Custom-trained dedicated YOLOv8 models per violation on **10,000+ Roboflow-annotated images** via Google Colab, achieving **~0.91 mAP@0.5** (93% precision, 89% recall) over a generic baseline.
- Engineered an adaptive OpenCV preprocessing pipeline (CLAHE low-light, denoising, deblur, shadow/haze correction) and an automated evidence generator producing annotated images + JSON metadata stored in SQLite.
- Delivered a decoupled **FastAPI REST API** and multi-page **Streamlit dashboard** (live analysis, records, analytics, reports); Dockerized and deployed live on Hugging Face Spaces.

CollabBoard — Full-Stack Real-Time Collaboration Platform

🐙 GitHub | May 2026

Next.js, React, Express.js, Socket.IO, PostgreSQL, Prisma, Tailwind CSS, Docker

- Engineered a full-stack workspace platform with a **Next.js** frontend and **Express.js** backend, enabling teams to manage projects, Kanban boards, and documents in a unified interface.
- Implemented live multi-user synchronization utilizing **WebSockets, Socket.IO, and CRDTs**, powering real-time chat and concurrent rich-text document editing.
- Architected a scalable relational database using **Prisma ORM and PostgreSQL** to handle complex data structures, user sessions, and granular role-based access control.
- Orchestrated decoupled deployments across Vercel and Render using **Docker**, engineering Next.js reverse proxy rewrites to bypass strict browser cross-domain tracking policies.

IECR — Indian Entertainment & Community Reviews

🐙 GitHub | June 2025

React.js, Node.js, Express.js, MongoDB, Socket.io, Clerk, TMDB API

- Architected a full-stack entertainment platform with **real-time chat, friend requests, and live notifications** using persistent WebSocket connections (Socket.io) on a split Vercel + Render deployment.
- Secured the application with **Clerk authentication and Role-Based Access Control (RBAC)**, with event-driven user sync to MongoDB via Inngest webhooks.
- Integrated the **TMDB API** to serve **275+ movies and 39+ web series** with dynamic poster art, cast details, trailer links, and an interactive watchlist and rating system.
- Hardened the backend with **Zod validation**, centralized error handling, structured logging (Pino), and a **Vitest unit test suite** for production-grade reliability.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C, C++, HTML, CSS

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, FastAPI, Streamlit, Tailwind CSS, Socket.IO

Databases: MongoDB Atlas, PostgreSQL, SQLite, MySQL

Tools & Platforms: Git, GitHub, Docker, Postman, Prisma, VS Code, Vercel, Figma, Render, Hugging Face

ACHIEVEMENTS

- Solved **750+ problems** on LeetCode & Codeforces, including medium/hard problems in **Dynamic Programming, Graphs, Trees, and Binary Search**.
- Achieved **Pupil** rank on **Codeforces** with a peak rating of **1225**.
- Shortlisted in **Top 300 teams** nationally in **Amazon HackOn 6.0** | Secured **Top 1600 teams** nationwide in **Flipkart Gridlock 2.0**.
- **98.6 percentile** in JEE Mains | **AIR 16,000** in JEE Advanced.

LEADERSHIP & EXTRACURRICULARS

- **Batch Representative**, IIITM Gwalior — Coordinated academic and extracurricular communications between students and administration for the 2023 batch.
- Calling Team Member, **Aurora'24** Cultural Fest — Managed outreach and coordination for one of IIITM's largest annual events.
- **Bronze Medalist & Coordinator, Inter-IIIT Carrom:** Managed tournament scheduling for Twaran'25 and secured a national 3rd place finish in Twaran'26.